Problem statement:
Which S&P 500 sector had the most dispersion among its constituents over the past year, and why does that matter for someone running a long/short book?

SP500 Dispersion Analysis:
I performed a cross-sectional std deviation as in weighted standard deviation of each sector and ranked them against each other. I also then looked at the best and worst stock in those sectors to glean whether the analysis is actually telling a story.

Sector	Cross-Sec Std (↑ = more dispersion)	N Constituents
XLK	+169.86%	73
XLC	+85.03%	23
XLI	+64.43%	79
XLB	+42.29%	26
XLV	+28.82%	58
XLF	+24.24%	76
XLP	+23.99%	36
XLY	+22.51%	48
XLR	+20.68%	31
XLE	+19.56%	22
XLU	+12.97%	31

L/S Spread Analysis:

Sector	Best Stock	Best Weight	Best Ret	Best Wtd Contrib	Worst Stock	Worst Weight	Worst Ret	Worst Wtd Contrib	Raw L/S Spread	Wtd L/S Spread
XLK	SNDK	0.012	+1220.60%	+14.23%	IT	0.001	-61.73%	-0.05%	+1282.33%	+14.28%
XLC	SATS	0.044	+364.75%	+15.89%	TTD	0.027	-60.26%	-1.63%	+425.01%	+17.52%
XLI	FIX	0.012	+317.88%	+3.73%	CPRT	0.005	-41.97%	-0.23%	+359.84%	+3.96%
XLB	ALB	0.028	+149.97%	+4.27%	DD	0.023	-38.84%	-0.90%	+188.81%	+5.17%
XLF	HOOD	0.009	+64.37%	+0.58%	FISV	0.004	-74.87%	-0.33%	+139.24%	+0.91%
XLY	TPR	0.007	+93.59%	+0.64%	LULU	0.004	-45.25%	-0.16%	+138.84%	+0.80%
XLV	MRNA	0.003	+87.04%	+0.29%	BAX	0.002	-50.62%	-0.09%	+137.66%	+0.38%
XLE	APA	0.008	+100.85%	+0.79%	OKE	0.032	-9.16%	-0.29%	+110.01%	+1.07%
XLP	CASY	0.010	+65.19%	+0.66%	CPB	0.003	-44.19%	-0.12%	+109.37%	+0.78%
XLR	HST	0.014	+32.69%	+0.47%	ARE	0.008	-49.07%	-0.37%	+81.76%	+0.84%
XLU	NRG	0.024	+48.32%	+1.15%	AWK	0.018	-7.42%	-0.13%	+55.74%	+1.28%

Conclusion:

XLK (S&P 500 Technology sector) appears to show high dispersion analytically and it is on par with what the market has experienced since 3rd quarter of last year when doubt started to pour into the tech sector for the following reasons:

- Doubt on profitability of enormous CAPEX on AI build out by hyperscalers due to slow adoption Appearance of circular financing scheme through financial engineering
- Release of open source AI models such as DeepSeek that challenged the compute power requirements.
- Causing further doubt on profitability

- AI models cannibalizing SaaS companies
- Private credit(largely private AI debt) destabilization and redemption pause by Blackstone

However, chip and memory producers like Sandisk(SNDK) continue to give the market hope that there is still much to see in AI. In other words a rotation out of software in AI to hardware which is why we are seeing SNDK's massive growth. All in all it can be argued both qualitatively and quantitatively XLK is going through high dispersion.

Why Does This Matters for a Long/Short Book?

Dispersion is volatility. Volatility is preferred because it creates opportunity for alpha by selectively picking stocks. For instance if a L/S PM were to foresee a memory chip shortage plus IT and SaaS being replaced by AI then he could have earned a weighted return of +14.23% -(-61.73%) = 75.96% for the period of 4/1/2025 and 3/31/2026 by going short on IT and long on SNDK.

In a low low dispersion sector the lack of movement makes it difficult to profit even if the direction is right. However, high dispersion, can be equally damaging if for example a momentum trade turns negative due to mean reversion.